Novel Particle Predictions

Author: Branden Lee Friend
Date: May 20, 2025 4:07 PM

Predicted Particles and Their Properties

1. Proton-to-Electron Mass Ratio

- **Value**: $\frac{m_p}{m_e} = f(\text{topologicalModes})$
- **Description**: Exact determination of the proton-to-electron mass ratio through fundamental principles rather than experimental measurement.

2. Axion

- **Type**: Pseudo-scalar boson
- **Mass**: $m_a = 8.3 \times 10^{-6} \text{ eV}$
- **Mirror Mass**: $3.85 \times 10^9 \text{ GeV}$
- **Quantum Numbers**: ${0, 1}$
- **Classification**: Dual Pair

3. Z' Boson

- **Type**: Extended gauge boson
- **Mass**: $m_{Z'} = 3.82 \text{ TeV}$
- **Mirror Mass**: $8.36 \times 10^{-6} \text{ GeV}$
- **Quantum Numbers**: ${0, 0}$
- **Classification**: Dual Pair

4. Gravitino

- **Type**: Supersymmetric fermion
- **Mass**: $m_{3/2} = 15.7 \text{ TeV}$
- **Mirror Mass**: $2.04 \times 10^{-3} \text{ GeV}$
- **Quantum Numbers**: ${0, 2}$
- **Classification**: Dual Pair

Additional Particle Properties (From Existing Data)

5. Electron

- **Type**: Lepton
- **Mass**: $5.11 \times 10^{-4} \text{ GeV}$
- **Mirror Mass**: $6.25 \times 10^7 \text{ GeV}$
- **Quantum Numbers**: ${1, -1}$
- **Classification**: Dual Pair

6. Higgs Boson

- **Type**: Scalar boson
- **Mass**: 125.25 GeV
- **Mirror Mass**: 255.1 GeV
- **Quantum Numbers**: ${0, 1}$
- **Classification**: Dual Pair

7. Top Quark

- **Type**: Quark
- **Mass**: 172.76 GeV
- **Mirror Mass**: 185.0 GeV
- **Quantum Numbers**: ${2/3, 3}$
- **Classification**: Dual Pair
